第二章 小结 Summary

1、本质：能量守恒与转换定律

进入系统的能量	-	离开系统的能量	=	系统能量的变化

第二章 小结(续1)

2、热一律表达式： 通用式

$$\dot{Q} = \mathrm{d}E_{\mathrm{cv}} / \delta\tau + \left(h + c^2/2 + gz\right)_{\mathrm{out}} \dot{m}_{\mathrm{out}} - \left(h + c^2/2 + gz\right)_{\mathrm{in}} \dot{m}_{\mathrm{in}} + \dot{W}_{\mathrm{net}}$$

$$\delta Q = \mathrm{d}E_{\mathrm{cv}} + \delta W_{\mathrm{net}} + \left(h + c^2/2 + gz\right)_{\mathrm{out}} \delta m_{\mathrm{out}} - \left(h + c^2/2 + gz\right)_{\mathrm{in}} \delta m_{\mathrm{in}}$$

第二章 小结(续2)

通用式

$$\dot{Q} = \mathrm{d}E_{\mathrm{cv}} / \delta\tau + \left(h + c^2/2 + gz\right)_{\mathrm{out}} \dot{m}_{\mathrm{out}} - \left(h + c^2/2 + gz\right)_{\mathrm{in}} \dot{m}_{\mathrm{in}} + \dot{W}_{\mathrm{net}}$$

稳流： $\mathrm{d}E_{\mathrm{cv}} / \delta\tau = 0$ $\quad \dot{m}_{\mathrm{out}} = \dot{m}_{\mathrm{in}} = \dot{m}$

$$q = \Delta h + \Delta c^2/2 + g\Delta z + w_s$$

$$q = \Delta h + w_t$$

第二章 小结(续3)

通用式

$$\dot{Q} = \mathrm{d}E_{\mathrm{cv}} / \delta\tau + \left(h + c^2/2 + gz\right)_{\mathrm{out}} \dot{m}_{\mathrm{out}} - \left(h + c^2/2 + gz\right)_{\mathrm{in}} \dot{m}_{\mathrm{in}} + \dot{W}_{\mathrm{net}}$$

闭口系： $\dot{m}_{\mathrm{out}} = \dot{m}_{\mathrm{in}} = 0$

$$\dot{Q} = \mathrm{d}E_{\mathrm{cv}} / \delta\tau + \dot{W}_{\mathrm{net}} \Rightarrow \delta Q = \mathrm{d}E + \delta W$$

第二章 小结(续4)

通用式

$$\dot{Q} = \mathrm{d}E_{\mathrm{cv}} / \delta\tau + \left(h + c^2/2 + gz\right)_{\mathrm{out}} \dot{m}_{\mathrm{out}} - \left(h + c^2/2 + gz\right)_{\mathrm{in}} \dot{m}_{\mathrm{in}} + \dot{W}_{\mathrm{net}}$$

循环： $\mathrm{d}E_{\mathrm{cv}} / \delta\tau = 0$ **out = in**

$$\oint \dot{Q} = \oint \dot{W}_{\mathrm{net}} \Rightarrow \oint \delta Q = \oint \delta W$$

第二章 小结(续5)

通用式

$$\dot{Q} = \mathrm{d}E_{\mathrm{cv}} / \delta\tau + \left(h + c^2/2 + gz\right)_{\mathrm{out}} \dot{m}_{\mathrm{out}} - \left(h + c^2/2 + gz\right)_{\mathrm{in}} \dot{m}_{\mathrm{in}} + \dot{W}_{\mathrm{net}}$$

孤立系： $\dot{Q} = \dot{W}_{\mathrm{net}} = \dot{m}_{\mathrm{out}} = \dot{m}_{\mathrm{in}} = 0$

$$\Rightarrow \quad \mathrm{d}E_{iso} = 0$$

第二章 小结(续6)

3、热力学第一定律表达式和适用条件

$q = \Delta u + w$　闭口系统，任何过程

$q = \Delta u + \int p\mathrm{d}v$　简单可压缩闭口系统 准静态过程

$q = \Delta h + \Delta c^2/2 + g\Delta z + w_s$　开口系统 稳流过程

$q = \Delta h + w_t$ 或 $q = \Delta h + w_s$　忽略动、位能变化

第二章 小结(续7)

4、准静态下两个热力学微分关系式

$$\delta q = \mathrm{d}u + p\mathrm{d}v$$

$$\delta q = \mathrm{d}h - v\mathrm{d}p$$

适合于闭口系和稳流开口系

后续很多式子基于此两式

第二章 小结(续8)

5、*u*与 *h*

U, *H* 广延参数　*u*, *h* 比参数

U 系统本身具有的内部能量

H 不是系统本身具有的能量，
开口系中随工质流动而携带的、
取决于状态参数的能量

第二章 小结(续9)

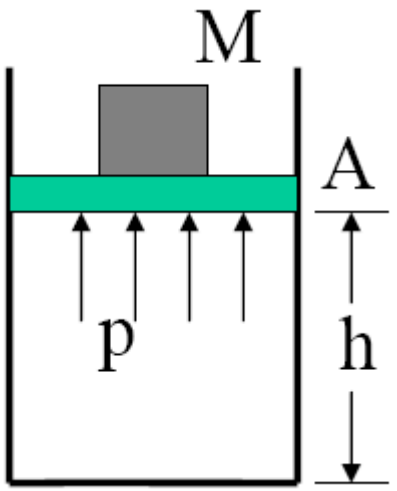
6、各种功及关系

容积变化功（闭口系准平衡过程）

$$\delta W = pA\,dh = p\,dV$$

$$\delta w = p\,dv$$

$$w = \int_1^2 p\,dv$$



第二章 小结(续10)

6、各种功及关系

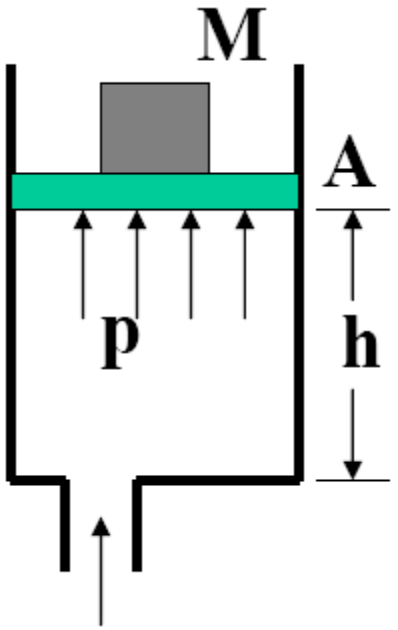
推挤功或推进功（开口系）

Push work

$$\delta W_{推} = pA\,dh = d\left(pV\right)$$

$$\delta w_{推} = d\left(pv\right)$$

$$w_{推} = pv$$

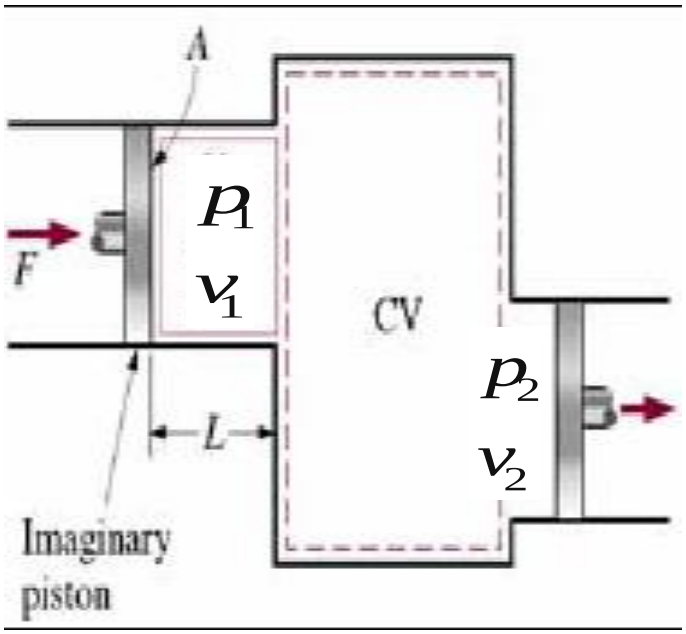


第二章 小结(续11)

6、各种功及关系

流动功（开口系）

Flow work



$$\delta w_f = d\left(p_2 v_2 - p_1 v_1\right)$$

$$w_f = p_2 v_2 - p_1 v_1 = \Delta\left(pv\right)$$

第二章 小结(续12)

6、各种功及关系

$w = \Delta(pv) + w_t$　做功的源泉（形式上）

$$w_t = \frac{1}{2}\Delta c^2 + g\Delta z + w_s$$

准静态过程　$w = \int p\mathrm{d}v$　闭口系

$w_t = -\int v\mathrm{d}p$　开口系

第二章 讨论课 Discussion

判断对误：

- 工质膨胀是否一定对外作功？

  做功对象和做功部件

- 定容过程是否一定不作功？

  开口系，技术功 $w_t = -\int v\mathrm{d}p$ 如水泵

  水轮机

第二章 讨论课(续1)

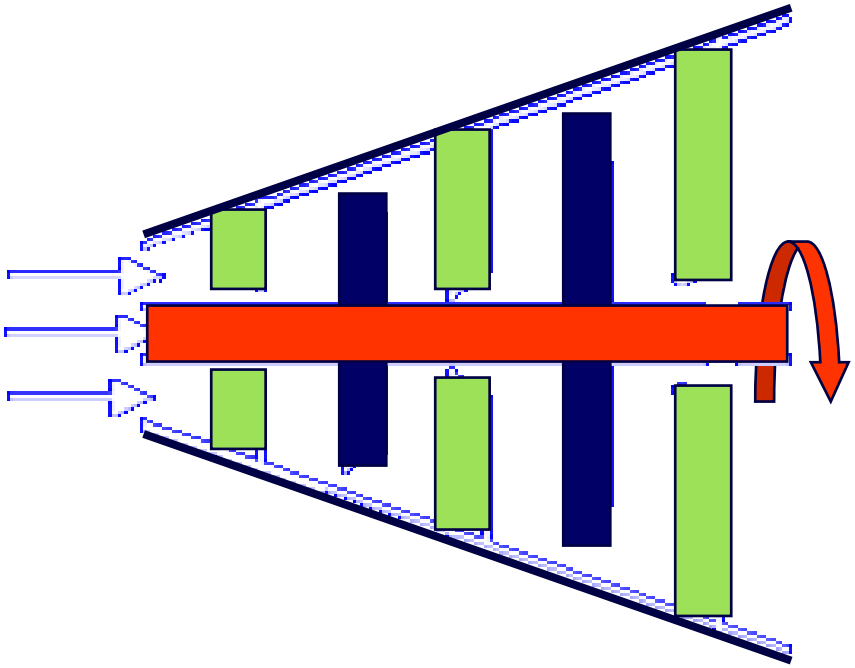
- 对工质加热，其温度反而降低，这种情况可能吗？

$$Q = \Delta U + W$$

$$>0 \quad <0 \quad >0$$

- 气体边膨胀边放热可能吗？

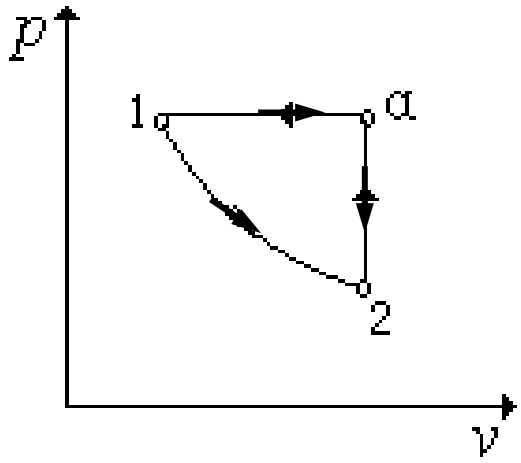
$$Q = \Delta H + W_t$$



第二章 讨论课(续2)

$W_{12} < W_{1a2}$　　$\Delta U_{12} = \Delta U_{1a2}$

$Q_{12} = \Delta U_{12} + W_{12}$　　$Q_{1a2} = \Delta U_{1a2} + W_{1a2}$



判断大小：　$Q_{12} < Q_{1a2}$

$W_{t12} < W_{t1a2}$　　$\Delta H_{12} = \Delta H_{1a2}$

$Q_{12} = \Delta H_{12} + W_{t12}$　　$Q_{1a2} = \Delta H_{1a2} + W_{t1a2}$　　$\oint Q =$

循环：　$\oint \Delta U = 0$　　$\oint \Delta H = 0$　　$\oint W = \oint W_t$

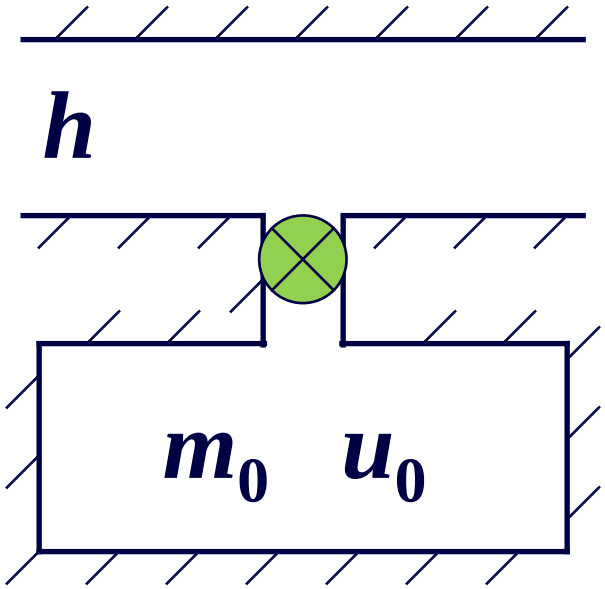
充气问题与取系统

储气罐原有气体 m_0，u_0

输气管状态不变，h不变

经τ时间充气，关阀

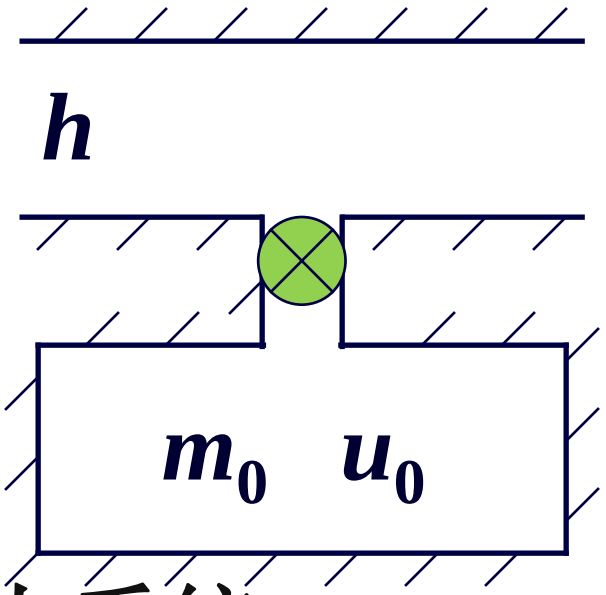
储气罐中气体m



求：储气罐中气体内能u'

忽略动、位能变化，且管路、储气罐、阀门均绝热

四种可取系统



1）取储气罐为系统

开口系

2）取最终罐中气体为系统

闭口系

3）取将进入储气罐的气体为系统

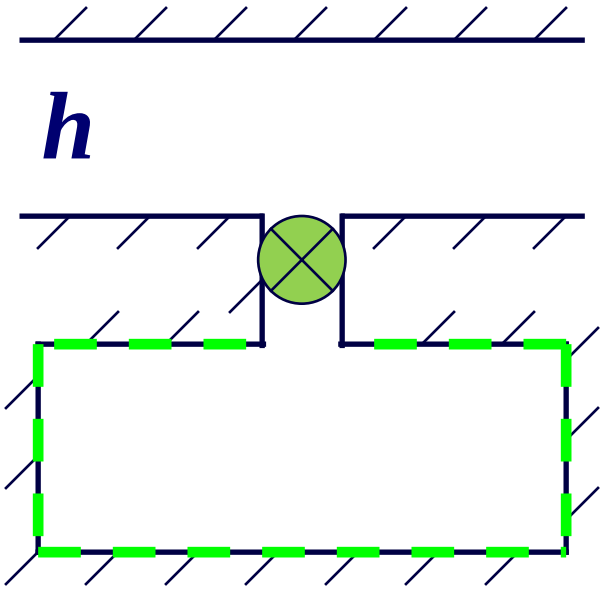
闭口系

4）取储气罐原有气体为系统

闭口系

1)取储气罐为系统(开口系)

$$\delta Q = \mathrm{d}E_{\mathrm{cv}} + \delta W_{\mathrm{net}} + \left(h + c^2/2 + gz\right)_{\mathrm{out}} \delta m_{\mathrm{out}} - \left(h + c^2/2 + gz\right)_{\mathrm{in}} \delta m_{\mathrm{in}}$$



忽略动位能变化

绝热

无作功部件

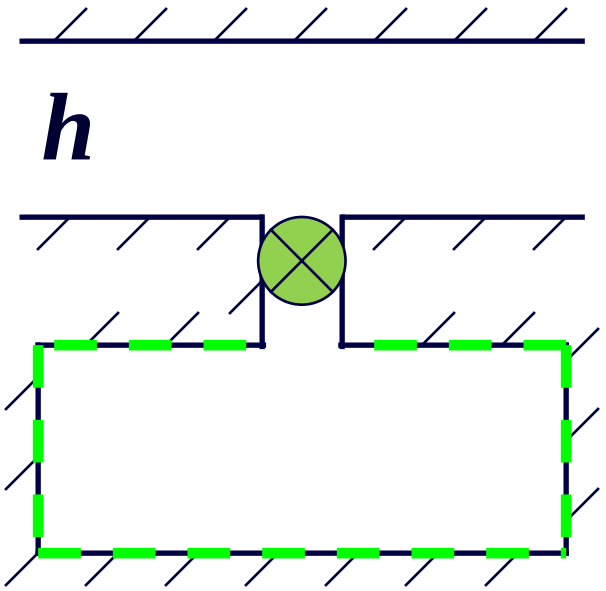
无离开气体

$$\mathrm{d}E_{\mathrm{cv}} - h\delta m_{\mathrm{in}} = 0$$

$$\mathrm{d}U_{\mathrm{cv}} = h\delta m_{\mathrm{in}}$$

1)取储气罐为系统(开口系)(续)

$$\mathrm{d}U_{cv} = h\delta m_{in}$$



经τ时间充气，积分概念

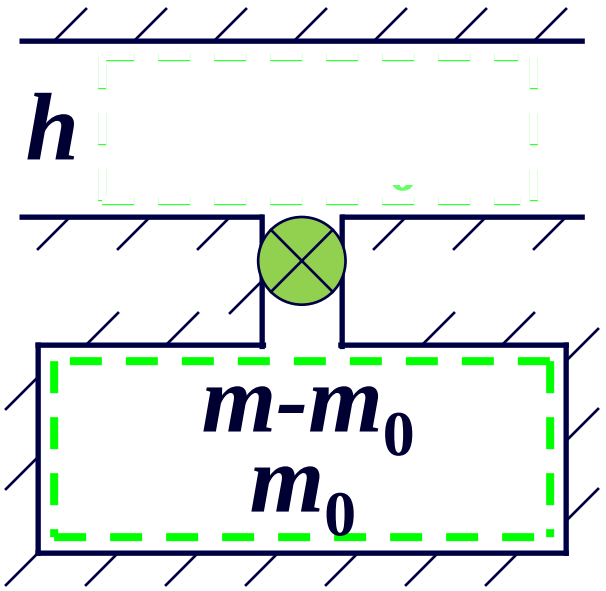
$$\int_{m_0u_0}^{mu'} \mathrm{d}U_{cv} = \int_0^{m-m_0} h\delta m_{in}$$

h是常数　$mu' - m_0u_0 = h(m - m_0)$

$$u' = \frac{h(m - m_0) + m_0u_0}{m}$$

2）取最终罐中气体为系统(闭口系)

$Q = \Delta U + W$ 绝热



$$\Delta U = m u' - \left[m_0 u_0 + (m - m_0) u \right]$$

$$W = -(m - m_0) p v$$

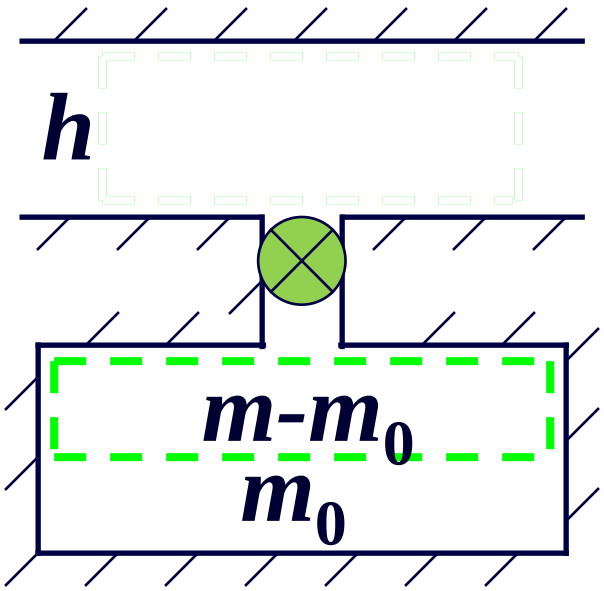
$$m u' - \left[m_0 u_0 + (m - m_0) u \right] - (m - m_0) p v = 0$$

$$m u' - m_0 u_0 - (m - m_0) h = 0$$

$$u' = \frac{h(m - m_0) + m_0 u_0}{m}$$

3)将进入储气罐的气体为系统(闭口系)

$$Q = \Delta U + W$$



m-m_0与m_0有温差传热 Q_1

$$\Delta U = (m - m_0)u' - (m - m_0)u$$

m-m_0对m_0作功W_1

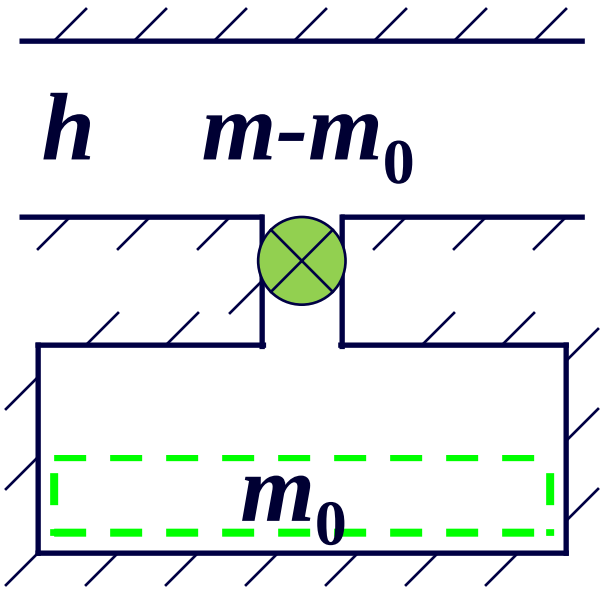
$$W = -(m - m_0)pv + W_1$$

$$Q_1 = (m - m_0)u' - (m - m_0)u - (m - m_0)pv + W_1$$

? $Q_1 = (m - m_0)u' - (m - m_0)h + W_1$?

4）取储气罐原有气体为系统(闭口系)

$$Q = \Delta U + W$$



m_0与$m-m_0$有温差传热Q_1'

$$\Delta U = m_0 u' - m_0 u_0$$

m_0得$m-m_0$作功W_1'

? $Q_1' = m_0 u' - m_0 u_0 + W_1'$?

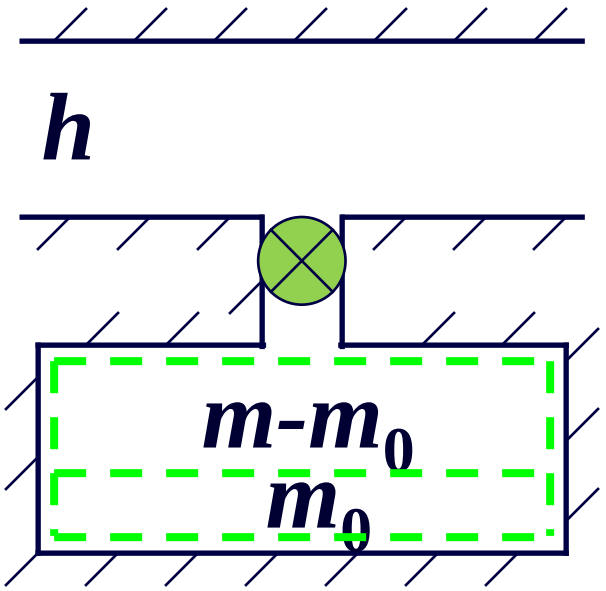
$$Q_1 = -Q_1' \qquad W_1 = -W_1'$$

$$Q_1 = (m - m_0)u' - (m - m_0)h + W_1$$

4)取储气罐原有气体为系统(闭口系)

$$Q_1' = m_0 u' - m_0 u_0 + W_1'$$

$$Q_1 = -Q_1' \qquad W_1 = -W_1'$$



$$Q_1 = (m - m_0)u' - (m - m_0)h + W_1$$

$$Q_1 - W_1 = -(Q_1' - W_1')$$

$$(m - m_0)u' - (m - m_0)h = -(m_0 u' - m_0 u_0)$$

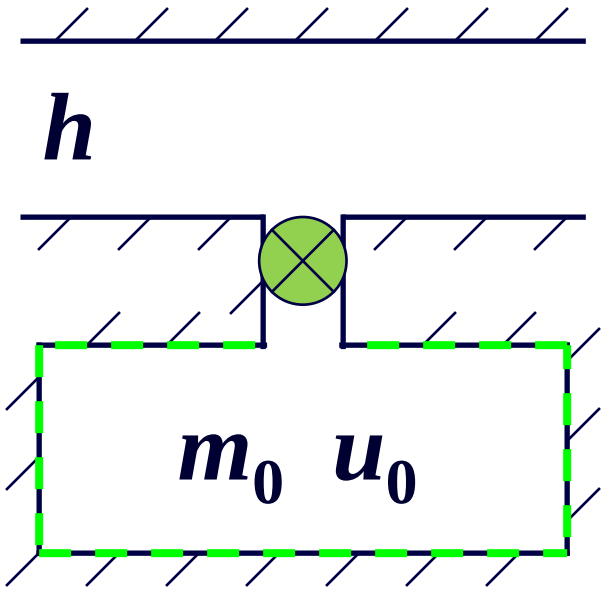
$$m u' = m_0 u_0 + (m - m_0)h$$

$$u' = \frac{h(m - m_0) + m_0 u_0}{m}$$

利用热一律的文字表达式

取储气罐为系统(开口系)

进 — 出 = 内能变化



进：$(m-m_0)h$

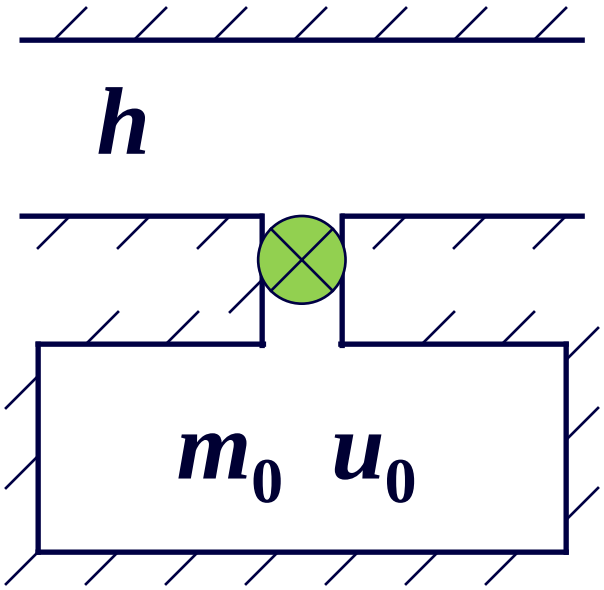
出：C

内能变化：$mu'-m_0u_0$

$$u'=\frac{h(m-m_0)+m_0u_0}{m}$$

总结

1）系统不同，但结果相同
取得巧，易处理



开口系反映为质量携带焓

闭口系反映为作功

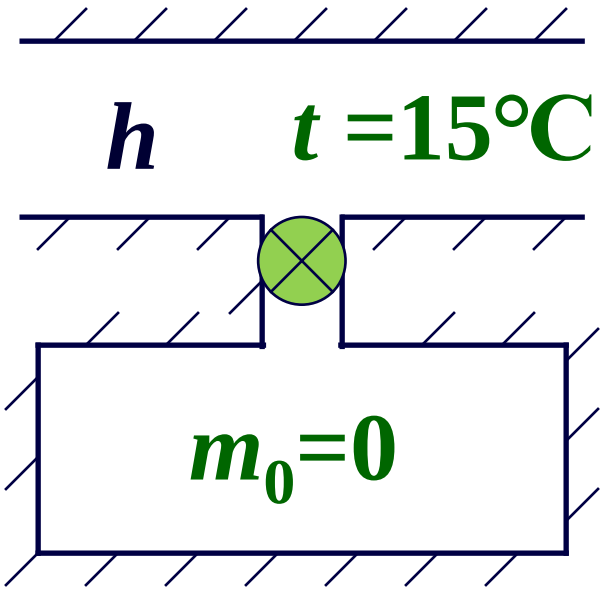
2）若$m_0=0$，

$$u' = h \qquad u' = \frac{h(m-m_0)+m_0u_0}{m}$$

充气的终温（$u' = h$ 时）

已知：理想气体　$u = c_v T$

$c_p / c_v = 1.4$　　$h = c_p T$



求：储气罐中气体终温

两种算法

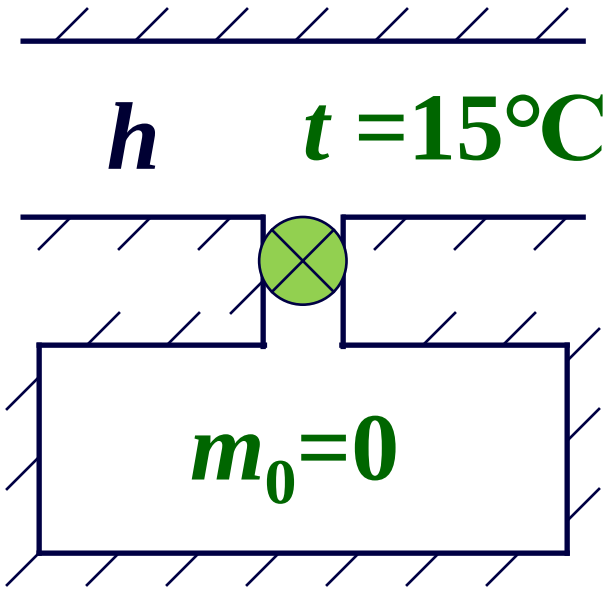
$c_v t' = c_p t \xrightarrow{℃} t' = (c_p / c_v)t = 21^o C$　　？

$c_v T' = c_p T \xrightarrow{K} T' = (c_p / c_v)T = 130.2^o C$

充气的终温（$u'=h$ 时）(续)

关键看u与h的零点
是否相同

$$h = u + pv = u + RT$$



理想气体:

$$pV = mRT \qquad pv = RT$$

0℃ $u = 0$ $h = R(273.15 + 0) \neq 0$

0K $u = 0$ $h = 0$

排气问题与取系统

已知：p_1=35bar, t_1 =16℃

$$u = c_v T \qquad h = c_p T$$

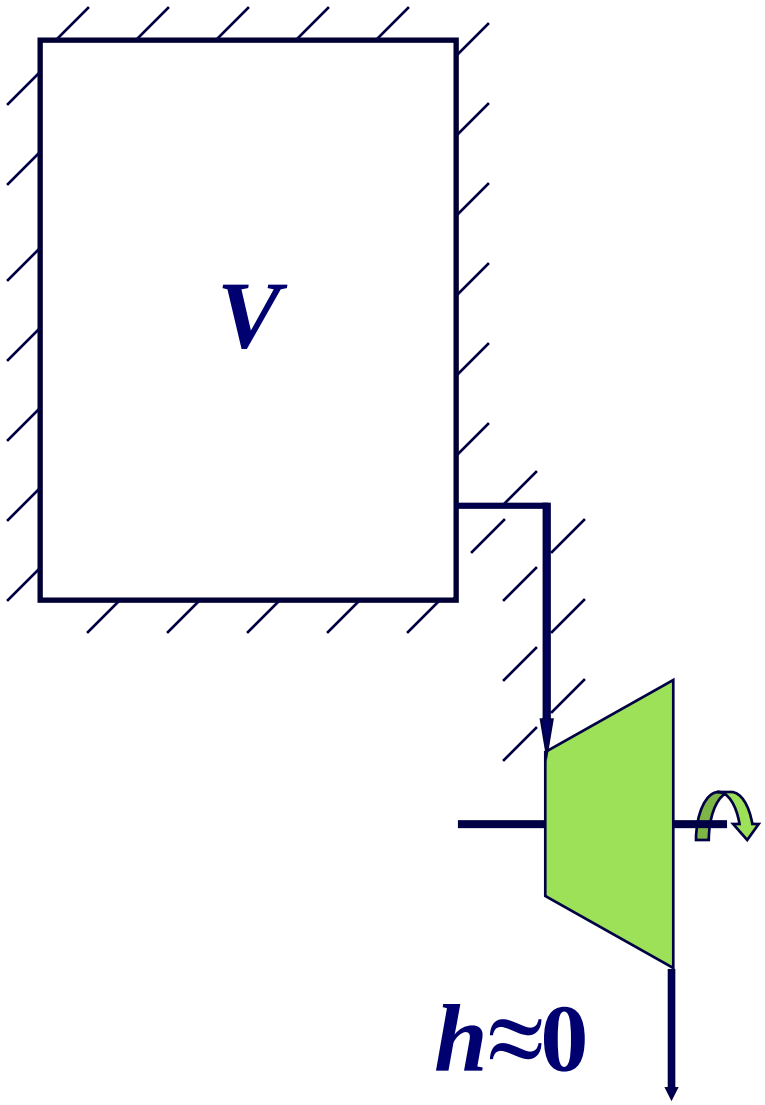
$$c_v = 718\ \mathrm{J/kg.K}$$

$$R = 287\ \mathrm{J/kg.K}$$

要求：输出4kW, 持续30s

允许：p_1 ⇩ p_2=3.5bar

求：需要的容积V



解1:取储气罐为系统（开口）

$$\delta Q = \mathrm{d}E_{\mathrm{cv}} + \delta W_{\mathrm{net}}$$

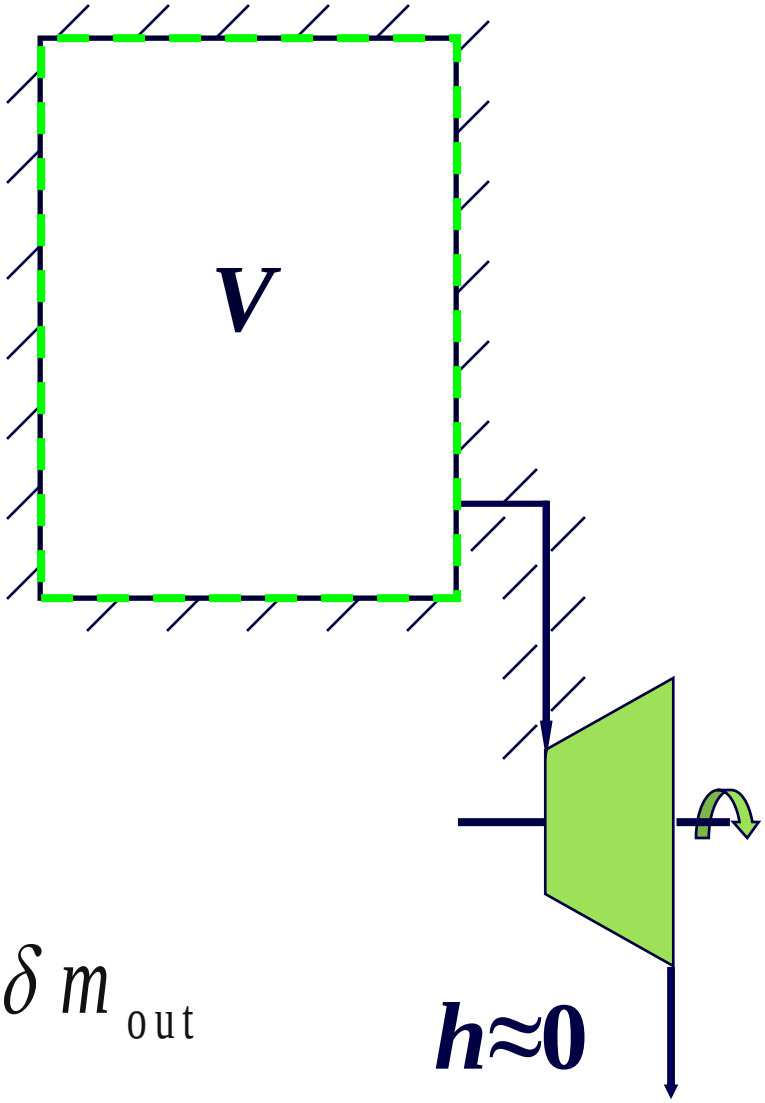
$$+\left(h + c^2/2 + gz\right)_{\mathrm{out}} \delta m_{\mathrm{out}}$$

$$-\left(h + c^2/2 + gz\right)_{\mathrm{in}} \delta m_{\mathrm{in}}$$

$$\mathrm{d}E_{\mathrm{cv}} + h\delta m_{\mathrm{out}} = 0$$

$$U_2 - U_1 = -\int h\delta m_{\mathrm{out}}$$

$$U_1 - U_2 = \int h\delta m_{\mathrm{out}}$$

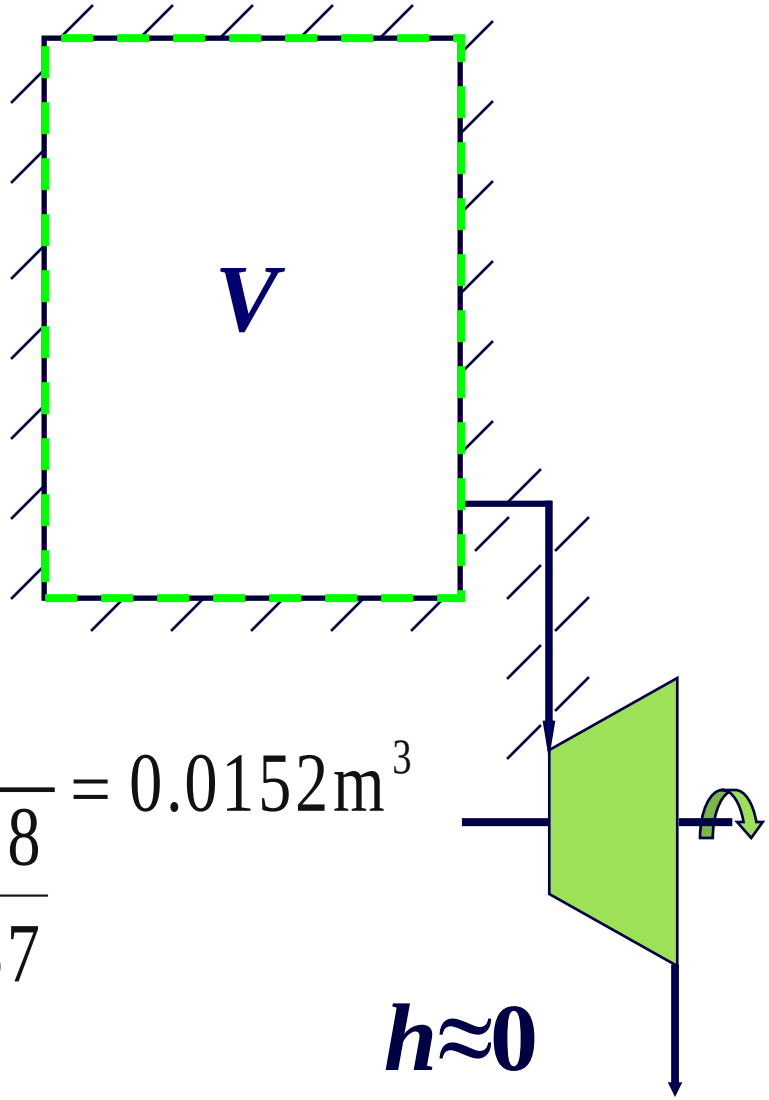


解1:取储气罐为系统（开口）

$$U_1 - U_2 = \int h\delta m_{out} \quad W$$

$$m_1 u_1 - m_2 u_2 = W$$

$$\frac{p_1 V}{RT_1} c_v T_1 - \frac{p_2 V}{RT_2} c_v T_2 = W$$



$$V = \frac{W}{(p_1 - p_2)\frac{c_v}{R}} = \frac{4 \times 30 \times 10^3}{(35 - 3.5) \times 10^5 \frac{718}{287}} = 0.0152 \mathrm{m}^3$$

解2：取储气罐和汽机为系统（开口）

进 — 出 = 内能变化

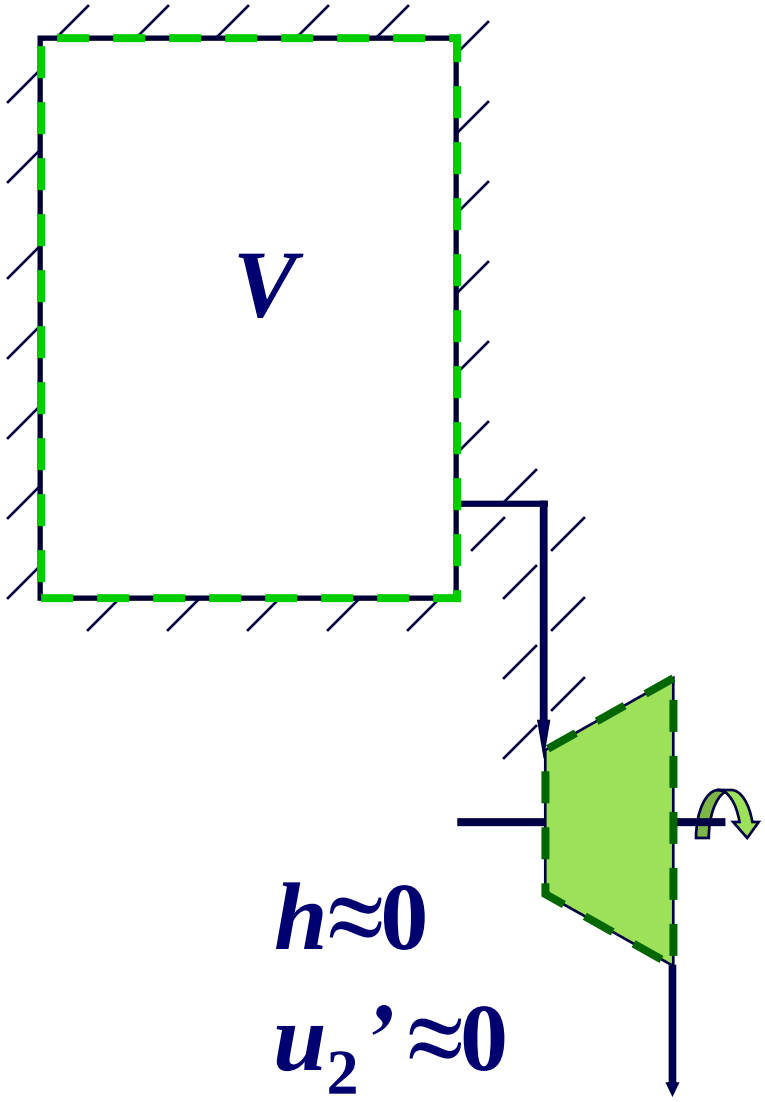
进： C

出： W

内能变化： $m_2u_2 - m_1u_1$

$$0 - W = m_2u_2 - m_1u_1$$

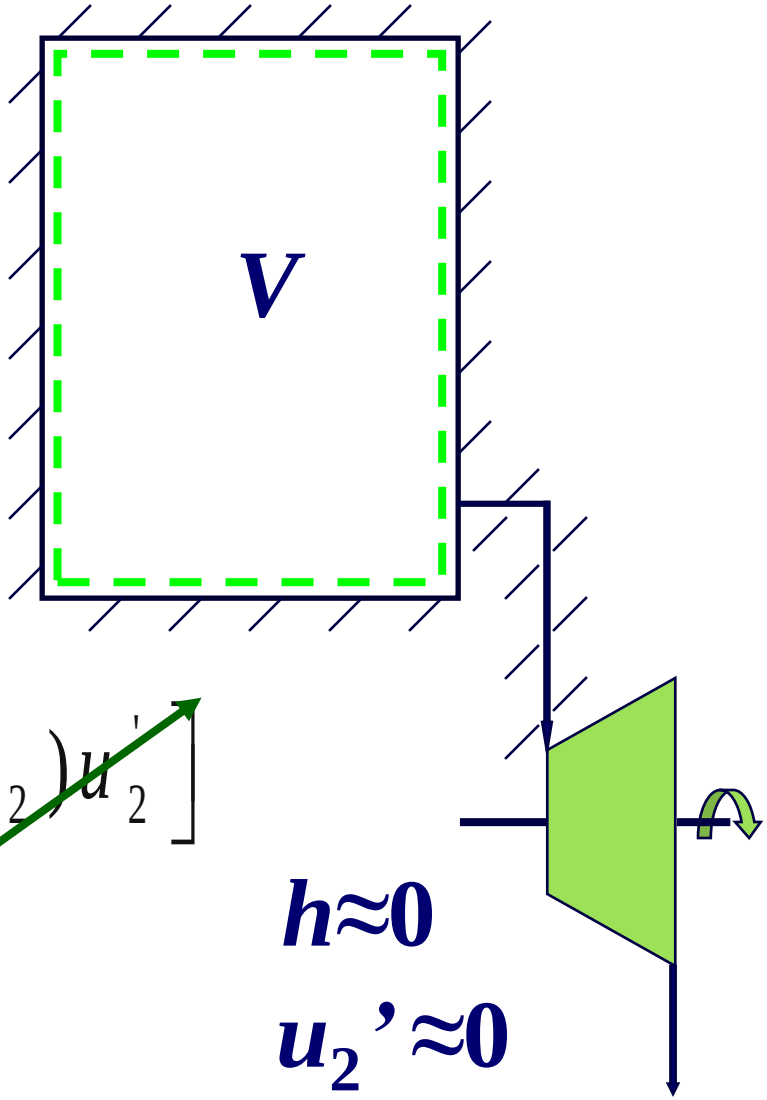
$$W = m_1u_1 - m_2u_2$$



解3：取气体为系统（闭口）

$$Q = \Delta U + W$$

$$W = U_1 - U_2$$



$$W = m_1 u_1 - \left[m_2 u_2 + (m_1 - m_2) u_2' \right]$$

$$W = m_1 u_1 - m_2 u_2$$

其它算功例

已知：缓慢加热He气

p_1=1.013bar, p_2 =3.039bar

活塞面积A=0.1m^2,无摩擦

弹簧刚度k=10^{5}N/m



求：He作功量W

物理学过：弹簧变形与力的关系 $\tau = kl$

弹簧功 $W_{\text{弹簧}} = -\int_{l_1}^{l_2} \tau \,\mathrm{d}l = -k\int_{l_1}^{l_2} l\,\mathrm{d}l = -\frac{1}{2}kl^2 \uparrow_{l_1}^{l_2}$

解1：由弹簧功求He作功量

$$l_1 = \frac{\tau_1}{k} = \frac{p_1 A}{k} = 0.1013\text{m}$$

$$l_2 = \frac{\tau_2}{k} = \frac{p_2 A}{k} = 0.3039\text{m}$$



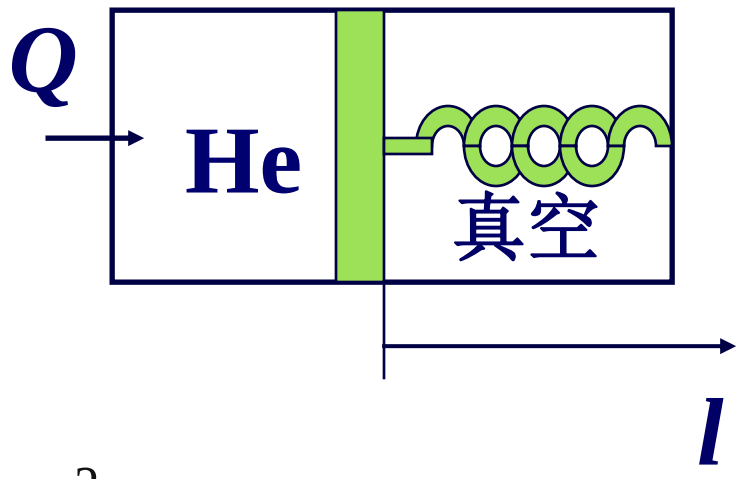
$$W_{\text{弹簧}} = -\int_{l_1}^{l_2} \tau \,\mathrm{d}l = -k\int_{l_1}^{l_2} l\,\mathrm{d}l = -\frac{1}{2}kl^2 \uparrow_{l_1}^{l_2}$$

$$W_{\text{弹簧}} = -4.105\ \text{kJ}$$

$$W_{\text{He}} = 4.105\ \text{kJ}$$

解2：由He参数求作功量

缓慢加热：准静态



$$W_{\mathrm{He}} = \int p\,\mathrm{d}V$$

***p* 与*V* 的关系？**

$$\frac{\mathrm{d}V}{A} = \mathrm{d}l = \frac{A\,\mathrm{d}p}{k} \qquad \mathrm{d}V = \frac{A^2\,\mathrm{d}p}{k}$$

$$W_{\mathrm{He}} = \int_{p_1}^{p_2} p\,\frac{A^2}{k}\,\mathrm{d}p = \frac{A^2}{k}\,\frac{p^2}{2}\uparrow_{p_1}^{p_2} = 4.105\ \mathrm{kJ}$$